

1 Neural Network Features for Kernel Methods

1.1 Theory

1.1.1 (a) Kernel Methods

Valid Kernel Function

A kernel function is a function

$$K(x, x') : \mathcal{X} \times \mathcal{X} \rightarrow R$$

that measures the similarity between two inputs x and x' .

A kernel is said to be **valid** if it can be expressed as the inner product of two vectors in some feature space:

$$K(x, x') = \langle \phi(x), \phi(x') \rangle$$

where

$$\phi : \mathcal{X} \rightarrow \mathcal{H}$$

is a mapping from the input space to a (possibly high-dimensional) feature space $\mathcal{H}$.

Common examples of kernel functions include:

Linear Kernel

$$K(x, x') = x^T x'$$

Polynomial Kernel

$$K(x, x') = (x^T x' + c)^d$$

Radial Basis Function (RBF) Kernel

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right)$$

Mercer's Theorem

Mercer's theorem states that if a kernel function $K(x, x')$ is continuous, symmetric, and positive semi-definite, then it can be expressed as:

$$K(x, x') = \sum_{i=1}^{\infty} \lambda_i \psi_i(x) \psi_i(x')$$

where

- $\lambda_i \geq 0$ are eigenvalues
- $\psi_i(x)$ are orthogonal eigenfunctions

This representation implies the existence of a feature mapping

$$\phi(x) = [\sqrt{\lambda_1}\psi_1(x), \sqrt{\lambda_2}\psi_2(x), \dots]$$

such that

$$K(x, x') = \phi(x)^T \phi(x')$$

Thus, kernel methods implicitly compute inner products in a high-dimensional feature space without explicitly computing $\phi(x)$.

1.1.2 (b) Neural Networks as Feature Representations

Consider a neural network with one hidden layer:

$$h = \sigma(Wx + b)$$

where

- x is the input vector
- W is the weight matrix
- b is the bias vector
- σ is the activation function
- h is the hidden layer output

The hidden layer can be viewed as a feature transformation:

$$\phi(x) = h = \sigma(Wx + b)$$

Thus, the neural network transforms the input x into a new feature representation $\phi(x)$.

The output layer then performs a linear operation on these features:

$$y = v^T \phi(x)$$

where v represents the output weights.

Relation to the Kernel Trick

Kernel methods also rely on feature mappings of the form:

$$K(x, x') = \phi(x)^T \phi(x')$$

The key difference is:

- Kernel methods use a predefined feature mapping $\phi(x)$ that is usually implicit.
- Neural networks learn the feature mapping $\phi(x)$ from the data through training.

Therefore, the hidden layers of a neural network can be interpreted as **learned feature extractors**. If we compute the inner product of these learned features,

$$K(x, x') = \phi(x)^T \phi(x')$$

we obtain a kernel induced by the neural network representation. In this sense, neural networks provide adaptive feature mappings similar to those used in kernel methods.